

Bulut Karakaya

bulutk@gmail.com | www.bulutk.com | Cambridge, UK

Summary

Game technology leader and studio founder with 16+ years shipping award-winning mobile and PC/console titles. Built and led teams on Monument Valley 3, Desta, and Star Trek: Fleet Command. Experienced in Unity and Unreal, focused on gameplay systems, AI, and technical direction. Currently building AI-assisted game systems and LLM-driven simulation for new forms of gameplay and player agency.

Work Experience

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| Sep 2025 – Present | Co-founder – Second River Games <ul style="list-style-type: none">Co-founded a UK studio making simulation and management games. Building Village of Whispers, where LLM-driven villagers interpret, remember, and gossip about player speech, over a traditional systems layer that keeps every emergent outcome traceable to a cause. |
| Apr 2025 – Sep 2025 | Head of Game Development – PhilosopherKing <ul style="list-style-type: none">Re-architected a server-bound prototype into a real-time Unity-based game engine, consolidating gameplay, AI, and data systems.Led and mentored a cross-discipline team, including 5 engineers transitioning from web to game development, helping establish ownership of gameplay features.Unified gameplay, AI, and UI pipelines, reducing iteration time from weeks to days.Defined studio-wide standards for AI-assisted programming workflows and build processes.Partnered with founders to prototype LLM-driven systems for adaptive characters and player-authored narrative experiences. |
| Mar 2021 – Apr 2025 | Lead Programmer – Ustwo Games, UK
Monument Valley 3 <ul style="list-style-type: none">Took over as Lead Programmer and overall technical lead mid-production, leading a 6-engineer team and aligning a cross-discipline group around a clear shipping focus.Facilitated key technical decisions across graphics, performance, and content pipelines, balancing expert input with production constraints.Oversaw integration with an external DLC partner, maintaining core system stability while enabling parallel content creation.Led the team through production to release, ensuring build stability and a smooth delivery. Desta: The Memories Between <ul style="list-style-type: none">Led the design and implementation of a new gameplay architecture in Unity, improving performance and simplifying integration across AI, UI and content systems.Reorganised a 3-engineer programming team, rebalancing responsibilities and hiring a senior systems engineer to strengthen technical depth and ownership.Built flexible AI and gameplay tools that allowed designers to author and iterate on behaviour without engineering support.Established clear production milestones and QA gates that enabled the team to ship without crunch, maintaining stability and quality through release. |
| Jul 2018 – Mar 2021 | Senior Gameplay Programmer – Jagex Ltd, Cambridge, UK <ul style="list-style-type: none">Supported the studio's transition from Unity to Unreal Engine 4, helping establish the initial project setup, player systems, and combat foundation.Implemented multiplayer combat and abilities with authoritative server logic, client synchronisation, and responsive player controls.Collaborated with design on the AI Director system (inspired by Left 4 Dead), which dynamically balanced pacing and difficulty and was later featured in a GDC talk by the designer.Shared Unreal knowledge and workflows across the gameplay team, contributing to a smoother technical transition and faster iteration. |
| Oct 2016 – Jun 2018 | Lead Unity Programmer – Digit Game Studio, Dublin, Ireland <ul style="list-style-type: none">Lead Programmer on Star Trek: Fleet Command (iOS / Android, Unity C#), guiding two engineers and preparing technical plans, estimates, and milestone schedules.Designed dynamic loading and memory-management systems, optimising scene transitions and reducing load times across devices.Collaborated with backend engineers to introduce real-time client-server communication via sockets, replacing stateless polling and improving responsiveness. |
| Nov 2014 – May 2016 | Co-founder / Technical Director – Pera Games, Istanbul, Turkey <ul style="list-style-type: none">Co-founded the studio and raised £1M in seed funding (≈ \$500 K) through a successful investor partnership and Kickstarter campaign, building a nine-person team.Designed and programmed Overfall, a Unity-based rogue-like RPG for Steam featuring an ability system that enabled designers to create hundreds of new abilities without engineering support.Released Overfall to strong community reception and later showcased it at Unite 2016 as a featured game. |

Mar 2014 – Sep 2014	Project Partner – GriPati Digital Entertainment, Istanbul, Turkey Co-developed Egg Fight (iOS / Android, Unity C#), a casual fighting game with adaptive AI.
Aug 2012 – Feb 2014	Co-founder / Technical Director – Tart Games, Istanbul, Turkey - Co-founded the studio after incubation within Startup Kitchen, developing Kixel, a 3D real-time multiplayer football game for Facebook. - Designed and programmed core gameplay and real-time multiplayer systems. - Reached over one million players in the first year.
Sep 2010 – Dec 2011	Gameplay Programmer – Gravi, Istanbul, Turkey Developed interactive mini-games for Gravi Floor, a large-scale motion-tracked playground combining projection and physical interaction.
Jun 2009 – Aug 2009	Prototype Intern – Stupid Fun Club, Berkeley, USA Built prototype experiences for Will Wright's experimental entertainment lab exploring new forms of play at the intersection of games, technology, and storytelling.
Jul 2007 – Sep 2007	SDET Intern – Microsoft Corporation, Redmond, USA First intern from a Turkish university accepted to Microsoft HQ; contributed automated testing tools.

Selected Awards

London Games Festival 2025. Selected among eight industry leaders showcasing diverse talent and creativity

Monument Valley 3 **Best Mobile Game**, Develop:Star Awards 2025
Best UK Mobile Game, UKIE Awards 2025

Desta: The Memories Between **Best Mobile Game**, Develop:Star Awards 2023
Best UK Mobile Game, UKIE Awards 2024

Star Trek: Fleet Command **Strategy/Simulation Game**, Webby Awards 2019
Technical Achievement, Webby Awards 2019

Education

May 2010	Carnegie Mellon University - Entertainment Technologies Center (ETC) - Master Of Entertainment Technology - Created SurfaceScapes, a D&D experience on Microsoft Surface table. Presented in GDC'10 and PAX East. - Teaching Assistant for Building Virtual Worlds course.
May 2008	Sabanci University - Faculty of Engineering and Natural Sciences - BSc in Computer Science and Engineering - Teaching Assistant for 3D Modelling and 3D Animation courses.

Certificates

May 2024	Mental Health First Aid (MHFA England)
Oct 2020	Improving Deep Neural Networks - DeepLearning.AI (Coursera)
Oct 2020	Neural Networks and Deep Learning - DeepLearning.AI (Coursera)
Aug 2020	Machine Learning - Stanford University (Coursera)

Skills

Leadership & Production:	Technical direction, team mentoring & hiring, roadmap planning, agile development, cross-discipline collaboration
Systems:	Gameplay architecture, AI systems, LLM integration, procedural generation, performance optimisation
Pipelines & QA:	Code review workflows, build/release management, QA processes
Programming:	C#, C++
Engines:	Unity, Unreal Engine
Tools:	Git, Perforce
Language:	English (fluent), Turkish (Native)

Interests

Certified diver, Argentine Tango dancer & teacher, Wing-Tsun practitioner, snowboarder and aerial acrobat.